

Exp. No.: 2	Palletizing Operation
Date:	

Aim:

To program the robot to pick up elements from one pallet and place them into a desired pallet.

Procedure:

- Power On and Initialize the Robot
- Command the robot to move to its designated home position.
- Activate the gripper and ensure it is in the open position.
- Define the pallets for 9 elements being picked at (344, -590,z) and placed at (117, 423, z), with 75 mm pitch using the command 'Def Plt', the Zig-Zag direction may be selected.
- Program the robot to move to the calculated position for picking up the object.
- Command the gripper to close and securely grasp the object.
- Lift the object and move it to the desired location.
- Adjust the robot's trajectory to avoid obstacles or other robots in the workspace.
- Reach to the position over the target location.
- Command the gripper to open and release the object.
- Repeat the sequence until all the objects are picked in the desired position and placed in the target position.
- Send the robot back to its home position once the stacking operation is complete.

Def Plt (Define pallet) : [MELFA-BASIC VI]

[Function]

Defines the pallet. (3-point pallet, 4-point pallet)

[Format]

Def[] Plt[] <Pallet No.>, <Start Point>, <End Point A>, <End Point B>, [<Diagonal Point>], <Quantity A>, <Quantity B>, <Pallet pattern>

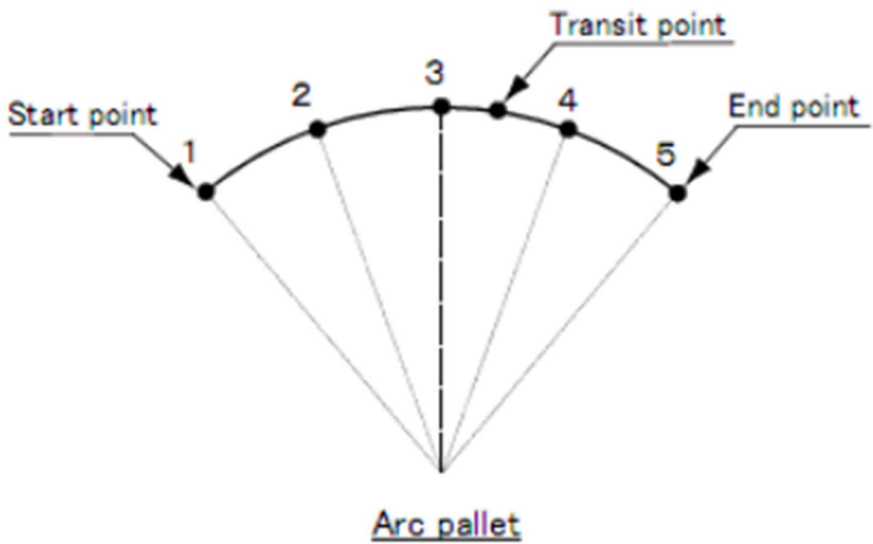
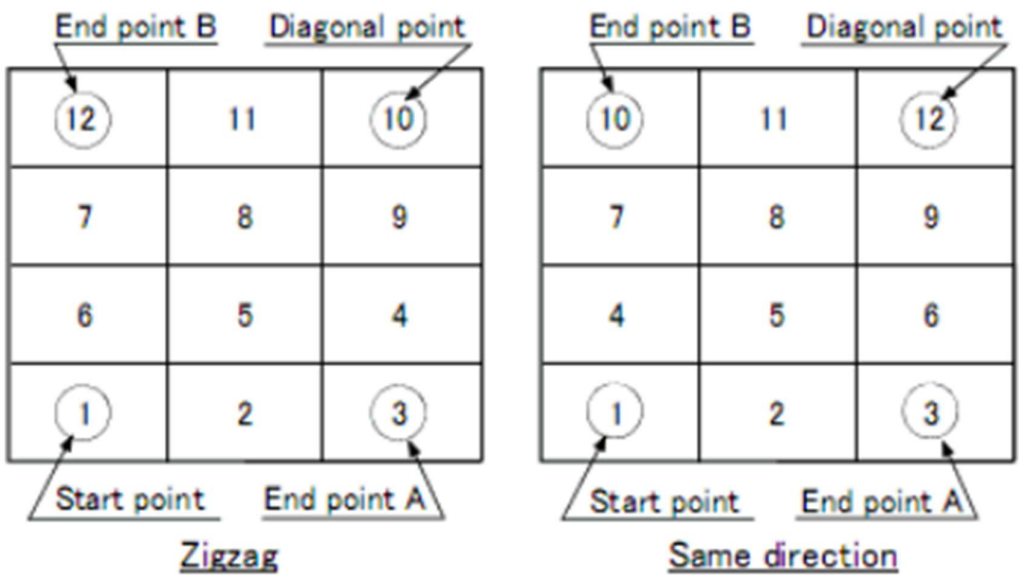
[Terminology]

<Pallet No.>	This is the selection No. of the set pallet. (Constants from 1 to 8 only).
<Start Point>	Refers to the pallet's start point.
<End Point A>	One of the ending points for the pallet. Transit point of arc for arc pallet.
<End Point B>	Another ending point for the pallet. Ending point of arc for arc pallet.
<Diagonal Point>	The diagonal point from the pallet's start point. Insignificant for arc pallet.
<Quantity A>	The No. of workpieces from the pallet's start point to the end point A. The No. of workpieces between the pallet start point and arc end point when using an arc pallet.
<Quantity B>	The No. of workpieces from the pallet's start point to the end point B. Insignificant for an arc pallet. (1, etc., must be designated.)

<Pallet Pattern> Specify the pallet pattern and fixation/equal division of the posture when numbering divided grid points. Constant only.

→ Types of Patterns (we use mostly)

1: Zigzag (posture equal division)	2: Same direction (posture equal division)
3: Arc pallet (posture equal division)	11: Zigzag (posture fixation)
12: Same direction (posture fixation)	13: Arc pallet (posture fixation)



RT Toolbox3 Program:

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Observation:

Analyze the recorded waypoints and gripper commands to ensure they accurately stand for the desired pick and place task.

Results & Inference: